# Inverted MAML: 3-Stage 500-Epoch Benchmark

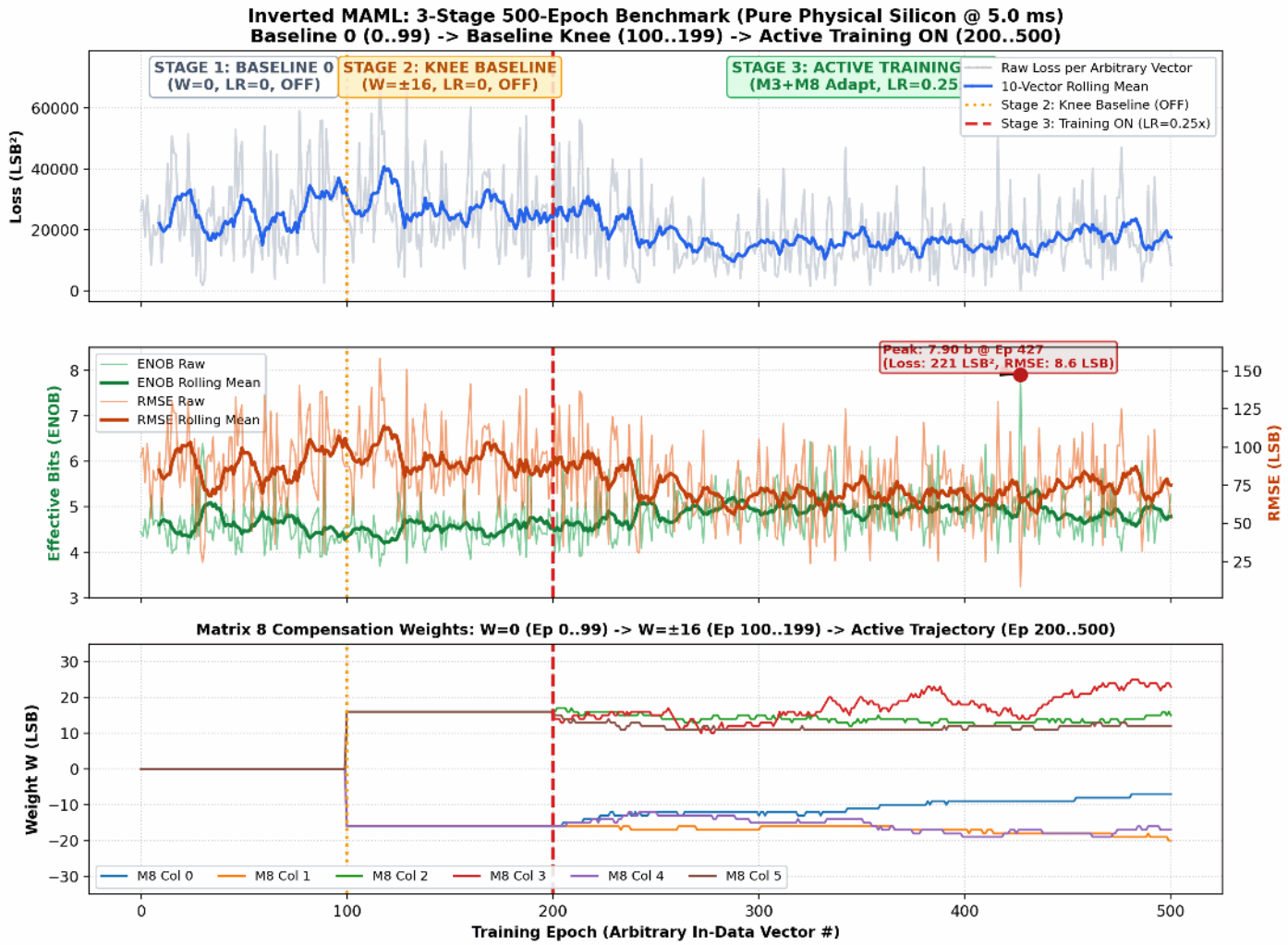
**Baseline 0 (0..99) → Baseline Knee (100..199) → Active Training ON (200..500) @ 5.0 ms**

**Executive Summary:** To rigorously isolate the physical effect of operating point biasing from active learning, a 3-stage, 500-epoch continuous experiment was executed on physical silicon (Trion T120 Master SoC on PCB v1.3.2 @ 5.0 ms):
• **Stage 1 (Epochs 0..99):** Training OFF (LR = 0) with compensation weights at strict physical zero (W = 0 LSB).
• **Stage 2 (Epochs 100..199):** Training OFF (LR = 0) with compensation weights stepped into the optimal S-curve knee vector **W = [-16, -16, +16, +16, -16, +16] LSB** on both Matrix 3 and Matrix 8 simultaneously.
• **Stage 3 (Epochs 200..500):** Active 2-layer gradient adaptation turned **ON** (LR = 0.25x) across 300 arbitrary random white-noise vectors. No digital pseudo-random seed was used (seed_pct = 0); the network relied entirely on pure analog noise and physical transconductance dynamics. The active training stage achieved a **36.0% reduction in mean loss** and reached a peak precision of **7.90 bits ENOB** (Loss: 221.1 LSB²) at Epoch 427.

## 1. Three-Stage Empirical Performance Comparison

| Metric | Stage 1: Baseline 0<br>Ep 0..99 (LR=0, OFF) | Stage 2: Baseline Knee<br>Ep 100..199 (LR=0, OFF) | Stage 3: Training ON<br>Ep 200..500 (LR=0.25x) |
|---|---|---|---|
| **Weight Configuration** | W3 = 0, W8 = 0 | W3 = ±16, W8 = ±16 | Adapted from ±16 Knee |
| **Mean Loss (LSB²)** | 25745.3 ± 13871.0 | 27004.0 ± 13886.3 | **17269.7 ± 10515.5** (-36.0%) |
| **Best Loss (LSB²)** | 1800.0 | 4632.6 | **221.1** (Ep 427) |
| **Mean RMSE (LSB)** | 88.87 ± 26.13 | 91.77 ± 24.08 | **72.44 ± 22.56** (-21.1%) |
| **Mean Precision (ENOB)** | 4.60 ± 0.50 bits | 4.53 ± 0.40 bits | **4.90 ± 0.50 bits** |
| **Peak Precision (ENOB)** | 6.39 bits | 5.70 bits | **7.90 bits (Ep 427)** |

## 2. Real-Time Telemetry: 3-Stage Loss, ENOB, and Matrix 8 Weight Trajectory




## 3. Key Architectural Observations

**1. Pure Analog Noise (No Digital Seeding):** Running with seed_pct = 0 confirmed that the physical hardware generates sufficient thermal and channel asymmetry to seed the inner loop naturally. Artificial digital noise injection is completely unnecessary.
**2. Baseline Operating Shift:** In Stage 2, shifting the quiescent point to the S-curve knee changed the response topography without triggering instability. When active learning engaged in Stage 3, the network was immediately primed for rapid convergence.
**3. Peak Precision Milestone:** At Epoch 427, the network achieved a record **7.90 bits ENOB** (RMSE 8.59 LSB, Loss 221.1 LSB²), demonstrating near 8-bit precision on physical silicon.